

GOOGLE CYBERSECURITY CERTIFICATE

SQL Security

Google Cybersecurity Professional Certificate · Study Guide

Six security-focused SQL queries against `log_in_attempts` and `employees` tables — using `AND`, `OR`, `NOT`, and `LIKE` with wildcards to investigate suspicious login activity and scope system updates to the right employees.

Track	Google Cybersecurity Professional Certificate
Course	Assets, Threats, and Vulnerabilities
Tables	<code>log_in_attempts</code> , <code>employees</code>
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1 Investigating Login Activity

1. After-hours failed logins

A potential incident occurred after business hours (after 18:00). Query for every failed login after that time.



AND — both conditions must hold

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00' AND success = FALSE;
```

2. Login activity around a specific incident date

A suspicious event occurred on 2022-05-09. Query for that day and the day before.



OR — either condition can hold

```
SELECT *  
FROM log_in_attempts  
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

3. Logins from outside Mexico

Login attempts from outside Mexico were flagged as suspicious for this organization's expected traffic pattern.



NOT LIKE + wildcard — exclude matching rows

```
SELECT *  
FROM log_in_attempts  
WHERE country NOT LIKE 'MEX%';
```

2 Scoping Employee System Updates

4. Marketing employees in the East building



AND + wildcard match on office

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East%';
```

5. Finance or Sales employees



OR across two departments

```
SELECT *  
FROM employees  
WHERE department = 'Finance' OR department = 'Sales';
```

6. All employees outside IT



NOT LIKE — exclude one department

```
SELECT *  
FROM employees  
WHERE department NOT LIKE 'Information Technology';
```

3 Quick Reference



SQL filter operators — quick reference

AND	both conditions must be true
OR	either condition can be true
NOT	exclude matching rows
LIKE '%'	pattern matching with a wildcard